

LAW-EnterpriseSOCLab

Microsoft Sentinel Security Operations Center (SOC) Laboratory

Project Report

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Project Overview

This project demonstrates the implementation of a foundational Security Operations Center (SOC) laboratory using Microsoft Sentinel. The laboratory was designed to simulate real-world security monitoring, log analysis, and detection engineering workflows within Microsoft Azure.

The project focuses on collecting Windows Security Events through Azure Monitor Agent (AMA), storing telemetry within Azure Log Analytics Workspace, and using Microsoft Sentinel to develop Analytics Rules, investigate security events, and perform log analysis using Kusto Query Language (KQL).

The laboratory serves as a practical introduction to cloud-native SIEM technologies and establishes a strong foundation for advanced detection engineering and threat hunting projects.

Technologies

- Microsoft Sentinel
- Microsoft Azure
- Azure Log Analytics Workspace
- Azure Monitor Agent (AMA)
- Windows Security Events
- Sysmon
- Kusto Query Language (KQL)
- MITRE ATT&CK Framework

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2. Project Overview

Modern organizations generate thousands of security events every day, making it difficult for security teams to manually monitor and investigate suspicious activities. Security Information and Event Management (SIEM) platforms address this challenge by collecting, analyzing, and correlating security telemetry from multiple data sources to improve threat detection and incident response.

LAW-EnterpriseSOCLab was developed to build a practical understanding of Microsoft Sentinel, Microsoft's cloud-native SIEM and Security Orchestration, Automation, and Response (SOAR) platform. The project focuses on implementing a foundational Security Operations Center (SOC) laboratory capable of collecting Windows Security Events, generating security alerts, and performing log investigations using Kusto Query Language (KQL).

The laboratory uses Azure Monitor Agent (AMA) to collect Windows Security Events and Sysmon telemetry from a Windows 11 system. The collected data is forwarded to Azure Log Analytics Workspace, where Microsoft Sentinel analyzes the telemetry and applies Analytics Rules to identify suspicious authentication activities.

The project demonstrates the complete workflow of a basic SOC environment, including data collection, log ingestion, analytics rule development, alert generation, log investigation, and validation of detection use cases. While the scope is intentionally focused on Microsoft Sentinel fundamentals, the laboratory provides practical experience with cloud-native security monitoring and establishes a strong foundation for advanced detection engineering and threat hunting.

This project was implemented entirely within a controlled laboratory environment for educational and research purposes.

3. Project Objectives

The primary objective of LAW-EnterpriseSOCLab was to develop a foundational understanding of Microsoft Sentinel by implementing a practical Security Operations Center (SOC) laboratory in Microsoft Azure.

The project was designed to simulate core SOC monitoring activities, including security event collection, log analysis, detection engineering, and alert investigation within a controlled environment.

The specific objectives of the project were:

- Deploy Microsoft Sentinel within Microsoft Azure.
- Configure Azure Log Analytics Workspace for centralized log storage.
- Install and configure Azure Monitor Agent (AMA) for data collection.
- Collect Windows Security Events and Sysmon telemetry from a Windows 11 system.
- Develop Analytics Rules to detect common authentication-related security events.
- Perform log analysis using Kusto Query Language (KQL).
- Validate detections by generating controlled authentication events.
- Understand Microsoft Sentinel incident generation and alert workflows.
- Apply MITRE ATT&CK concepts to security detections.
- Build practical experience with cloud-native SIEM technologies and Security Operations Center workflows.

The successful completion of these objectives provides a solid foundation for advanced Microsoft Sentinel implementations, detection engineering, and enterprise threat hunting projects.

4. Lab Architecture

The LAW-EnterpriseSOCLab architecture was designed to demonstrate the complete data flow within a Microsoft Sentinel Security Operations Center (SOC) environment. The laboratory follows a cloud-native security monitoring model, where Windows security telemetry is collected, transmitted to Microsoft Azure, and analyzed using Microsoft Sentinel.

The environment consists of a Windows 11 endpoint configured with Sysmon and Azure Monitor Agent (AMA). Windows Security Events and Sysmon telemetry are continuously collected from the endpoint and securely transmitted to Azure Log Analytics Workspace through Azure Monitor Agent.

Azure Log Analytics Workspace serves as the centralized repository for security telemetry. Microsoft Sentinel is connected to the workspace and continuously analyzes incoming events using Analytics Rules developed within the project.

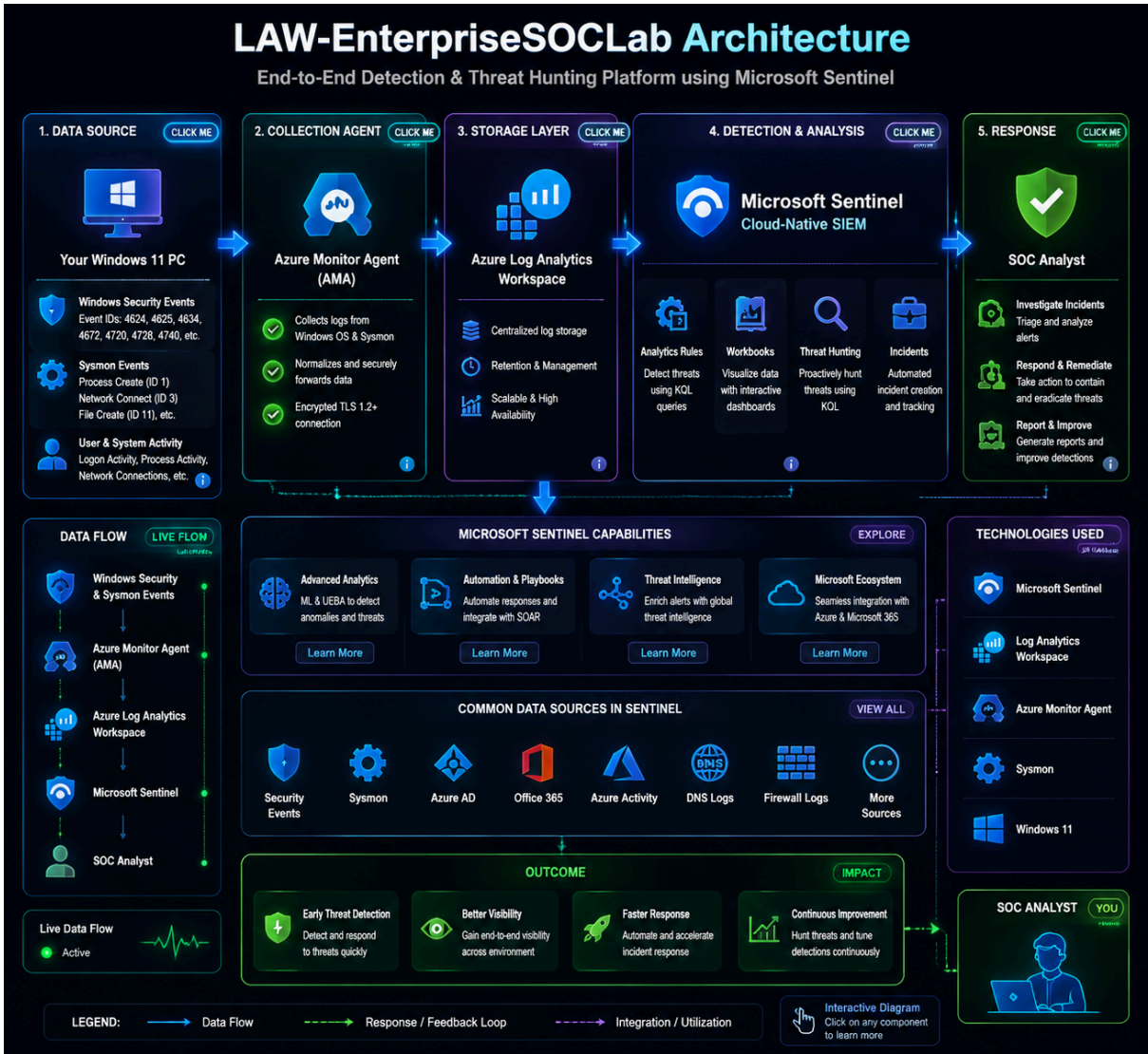
When predefined detection conditions are met, Microsoft Sentinel generates security alerts that can be reviewed by the analyst for further investigation. Security events are also queried using Kusto Query Language (KQL) to perform log analysis, validate detections, and investigate authentication-related activities.

The laboratory architecture demonstrates the fundamental workflow of a cloud-native SIEM platform, beginning with endpoint telemetry collection and ending with centralized monitoring, alert generation, and security investigation.

Figure 4.1 illustrates the overall laboratory architecture implemented during this project.

[Insert Architecture Diagram Here]

Figure 4.1 — LAW-EnterpriseSOCLab Architecture



5. Technology Stack

The LAW-EnterpriseSOCLab project utilizes Microsoft Azure security services and Windows telemetry sources to simulate a foundational Security Operations Center (SOC) environment. Each technology was selected to demonstrate a specific component of the security monitoring lifecycle.

Microsoft Sentinel

Microsoft Sentinel is the primary Security Information and Event Management (SIEM) platform used throughout the project. It provides centralized security monitoring, analytics rule management, alert generation, and log investigation capabilities using cloud-native services.

Microsoft Azure

Microsoft Azure provides the cloud infrastructure required to deploy Microsoft Sentinel and associated security services. It hosts the Log Analytics Workspace and enables centralized security monitoring.

Azure Log Analytics Workspace

Azure Log Analytics Workspace acts as the central repository for collected security telemetry. Windows Security Events and Sysmon logs are ingested into the workspace, where they become available for analysis using Kusto Query Language (KQL).

Azure Monitor Agent (AMA)

Azure Monitor Agent (AMA) is responsible for securely collecting and forwarding Windows Security Events and Sysmon telemetry from the Windows 11 endpoint to Azure Log Analytics Workspace.

Windows 11

Windows 11 serves as the monitored endpoint within the laboratory environment. Authentication events and operating system security logs generated by Windows are collected and analyzed through Microsoft Sentinel.

Sysmon

Sysmon extends native Windows logging by generating detailed endpoint telemetry, including process creation, network connections, and system activity. This additional visibility improves detection and investigation capabilities.

Kusto Query Language (KQL)

KQL is Microsoft's query language used to search, filter, correlate, and investigate security events stored within Azure Log Analytics Workspace. All log analysis and detection validation within the project were performed using KQL.

MITRE ATT&CK Framework

The MITRE ATT&CK Framework was used as a reference to understand how detection rules relate to common attacker tactics and techniques, helping align the laboratory with industry-recognized security practices.

6. Lab Environment & Deployment

The LAW-EnterpriseSOCLab environment was built using Microsoft Azure and a Windows 11 endpoint to demonstrate the deployment of a cloud-native Security Information and Event Management (SIEM) solution.

The deployment process began by creating a Log Analytics Workspace within Microsoft Azure. This workspace serves as the centralized repository for all security telemetry collected from the monitored endpoint.

Microsoft Sentinel was then enabled on the Log Analytics Workspace to provide security monitoring, analytics, investigation, and incident management capabilities.

A Windows 11 system was configured as the monitored endpoint. Sysmon was installed to generate enhanced endpoint telemetry, while Azure Monitor Agent (AMA) was deployed to securely collect Windows Security Events and Sysmon logs.

A Data Collection Rule (DCR) was configured to define which event logs would be collected and forwarded to Azure Log Analytics Workspace. Once the configuration was complete, Windows Security Events began flowing into Microsoft Sentinel, where they became available for analysis.

Analytics Rules were then created within Microsoft Sentinel to detect common authentication-related activities such as successful logons, failed logons, privileged logons, and account lockout events.

Finally, Kusto Query Language (KQL) was used to validate data ingestion, investigate collected events, and verify that the configured detections were functioning as expected.

The completed deployment provided a fully operational Microsoft Sentinel laboratory capable of collecting, analyzing, and investigating Windows security telemetry within a controlled environment.

7. Data Collection & Log Ingestion

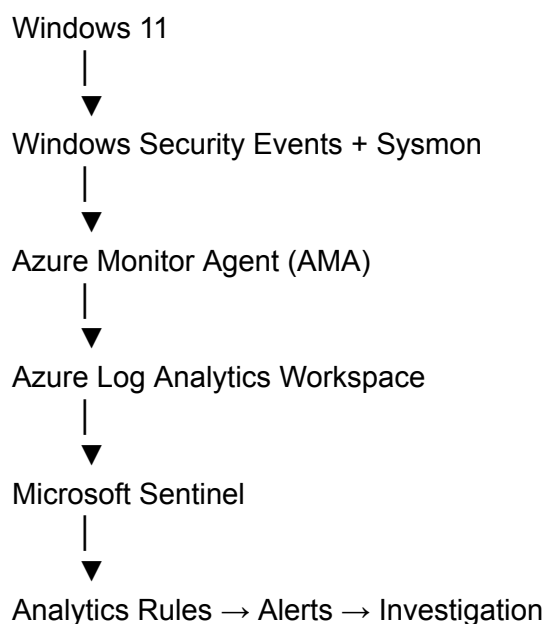
Effective security monitoring depends on reliable data collection. Within the LAW-EnterpriseSOCLab environment, Windows security telemetry is collected from the monitored endpoint and securely transmitted to Microsoft Sentinel through Azure Monitor Agent (AMA).

The Windows 11 endpoint continuously generates security events related to user authentication, account management, privilege usage, and other operating system activities. In addition to native Windows Security Events, Sysmon provides enhanced endpoint telemetry, offering greater visibility into system activity.

Azure Monitor Agent (AMA) is installed on the endpoint to securely collect this telemetry. Data Collection Rules (DCRs) define which event logs are collected and forwarded to Azure Log Analytics Workspace.

Once the telemetry reaches Azure Log Analytics Workspace, it becomes immediately available for analysis using Microsoft Sentinel. Security analysts can perform log searches using Kusto Query Language (KQL), create Analytics Rules, generate security alerts, and investigate suspicious activities from a centralized platform.

The overall data flow implemented in this laboratory is as follows:



This architecture demonstrates how Microsoft Sentinel receives endpoint telemetry, processes security events, and enables efficient monitoring and investigation within a cloud-native SOC environment.

8. Analytics Rules & Detection Engineering

Detection engineering is one of the primary functions of a Security Operations Center (SOC). It involves developing detection logic capable of identifying suspicious activities within collected security telemetry. Microsoft Sentinel provides Analytics Rules that continuously evaluate incoming events and generate alerts when predefined conditions are met.

Within the LAW-EnterpriseSOCLab environment, multiple Analytics Rules were created to monitor authentication-related events generated by the Windows 11 endpoint. These rules were developed using Kusto Query Language (KQL) and validated through controlled testing within the laboratory.

The implemented detections focused on common Windows authentication activities, enabling visibility into both normal and potentially suspicious user behavior.

The Analytics Rules implemented during this project include:

- Successful Logon Detection
- Failed Logon Detection
- Brute Force Detection
- Account Lockout Detection
- Privileged Logon Detection
- Suspicious Authentication Detection

Each Analytics Rule was configured with an appropriate severity level and mapped to relevant MITRE ATT&CK tactics where applicable. Once deployed, Microsoft Sentinel continuously monitored incoming telemetry and generated alerts whenever the configured detection conditions were satisfied.

The successful implementation of these Analytics Rules demonstrates a foundational understanding of detection engineering, alert generation, and security monitoring within Microsoft Sentinel.

9. KQL Queries & Log Investigation

Kusto Query Language (KQL) is the primary query language used within Microsoft Sentinel and Azure Log Analytics Workspace for searching, filtering, and analyzing security telemetry. Throughout this project, KQL was used to validate data ingestion, investigate authentication events, and verify that Analytics Rules were functioning correctly.

Several queries were executed to identify successful logons, failed logons, privileged logons, account lockout events, and generated security alerts. These queries enabled rapid investigation of collected telemetry and provided valuable insight into authentication activity occurring on the monitored Windows 11 system.

Example investigation activities performed during the project included:

- Viewing all collected security events
- Monitoring successful user logons
- Identifying failed authentication attempts
- Reviewing generated security alerts
- Validating Analytics Rule execution
- Investigating authentication-related events

Using KQL significantly improved the efficiency of log analysis by allowing specific events to be filtered, sorted, and correlated within Microsoft Sentinel. This capability forms the foundation of modern cloud-native Security Operations Center (SOC) investigations and enables security analysts to quickly identify suspicious activity.

The practical use of KQL throughout this laboratory strengthened understanding of Microsoft Sentinel log analysis and demonstrated how query-based investigations support security monitoring and incident response.

10. Project Results & Conclusion

The LAW-EnterpriseSOCLab project successfully demonstrated the implementation of a foundational Security Operations Center (SOC) laboratory using Microsoft Sentinel within Microsoft Azure.

Throughout the project, Windows Security Events and Sysmon telemetry were successfully collected from a Windows 11 system using Azure Monitor Agent (AMA). The collected telemetry was stored in Azure Log Analytics Workspace and analyzed through Microsoft Sentinel.

Multiple Analytics Rules were developed and validated to detect common authentication-related activities, including successful logons, failed logons, privileged logons, account lockouts, brute-force attempts, and suspicious authentication events. Kusto Query Language (KQL) was used to investigate collected logs, validate detections, and analyze generated security alerts.

Project Outcomes

- Successfully deployed Microsoft Sentinel.
- Configured Azure Log Analytics Workspace.
- Implemented Azure Monitor Agent (AMA).

- Collected Windows Security Events and Sysmon telemetry.
- Developed and validated Analytics Rules.
- Performed log investigation using Kusto Query Language (KQL).
- Generated and analyzed security alerts.
- Applied MITRE ATT&CK concepts to detection engineering.

The project provided practical experience with Microsoft Sentinel and strengthened understanding of cloud-native SIEM operations, detection engineering, and security monitoring.

LAW-EnterpriseSOCLab serves as a foundational Microsoft Sentinel project and establishes the knowledge required for more advanced SOC, detection engineering, and threat hunting implementations.